Computational Questions and Answers

1. Linear Regression

Model Definition:

Linear regression models the relationship between input variables (X) and a continuous target variable (y) using a linear predictor function.

$$y = Xw + b$$

Loss Function:

Mean Squared Error (MSE) is commonly used:

$$L(w, b) = (1/n) * \text{Sum } (y_i - (w^T x_i + b))^2$$

Gradient Descent:

Used to minimize the loss function by updating weights iteratively:

$$w := w - \alpha * dL/dw$$

$$b := b - \alpha * dL/db$$

where alpha is the learning rate.

Regularization:

Adds a penalty to the loss function to prevent overfitting.

L2 Regularization (Ridge):

L(w, b) = MSE + lambda * norm(w)^2

L1 Regularization (Lasso):

L(w, b) = MSE + lambda * norm(w)_1

2. Neural Network

Fully-Connected Layers:

Each neuron in a layer is connected to every neuron in the previous layer.

Forward Propagation:

Input is passed through the network layer-by-layer.

z = W*a_prev + b

a = activation(z)

Loss Function:

Cross-Entropy (for classification):

L = -Sum y_i * log(p_i)

Backward Propagation:

Calculates gradients of the loss function with respect to each parameter using the chain rule.

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dL/dW, dL/db computed layer by layer backward.

Parameters are updated via gradient descent.

w := w - alpha * dL/dw

b := b - alpha * dL/db

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